

Memory

Coming to the memory configuration, the NVIDIA cards use slightly faster GDDR6X memory whereas AMD uses GDDR6. Prima facie, this might look like an easy victory for NVIDIA but the fact is that memory type, the bus interface and the speeds are all determined as per the need of the GPU. Slapping on a faster memory type does nothing if the GPU was never bottlenecked to begin with. Despite the difference in memory types, the bandwidth available to both GPUs i.e. the RTX 4090 and the 7900 XTX is 1008 GB/s and 960 GB/s, respectively. So they've both got 24 GB of VRAM at nearly the same bandwidth. It comes down to how the GPU makes use of the available memory.

AI

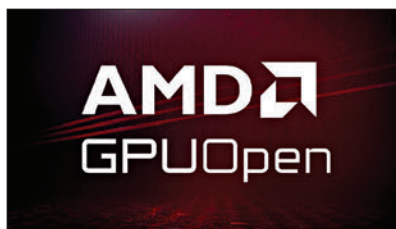
Digging a little deeper into the GPUs themselves, we come across a very similar hierarchy of arranging different computational units. There's the default render pipeline for raster workloads. Then there's the ray tracing aspect for which NVIDIA and AMD have specialised cores. NVIDIA has a higher RT core count than AMD and they're on the third generation so there's an obvious advantage. RT cores aside, NVIDIA also has Tensor Cores for AI which are currently in their fourth generation and you'd have to be living under a rock to not know about NVIDIA's market dominance for AI hardware. They became one of the most valuable companies on earth earlier this year because of how good their hardware is compared to all the alternatives. AMD is also gunning for the AI market with their new Instinct MI300 accelerators and NVIDIA has Blackwell coming out later this year. So the AI hardware market should see a slight change but NVIDIA will have the lead.

On the consumer side, not a lot of games are baking in AI features that would be compelling enough. As it stands, AI features are mostly quality-of-life improvements and not game changers. Most of the world still prefers raster gaming. The one thing that's making a change is the upscaling technologies. NVIDIA has DLSS and AMD has FSR and RSR. Both have ensured that gamers can

delay upgrades for a while purely because of how these upscaling technologies allow for higher frame rates by rendering smaller frames in larger quantities and upscaling them.

SOFTWARE

There are two aspects here - the drivers and the software ecosystem. NVIDIA comes across as a little protective of their IP with how they handle their drivers and how they're willing to collaborate with third-party members of the ecosystem. Linus Torvalds infamous rant in which he called NVIDIA "the single worst company" that the Linux Foundation had dealt with followed by an f-bomb drop, will forever live in the annals of the internet. That was back in 2012. NVIDIA didn't change anything until 2022 when it finally released the open-source GPU kernel modules with the R515 driver. It took them a decade but they finally came around to supporting Linux the proper way.



AMD, on the other hand, has been a champion of open-source for the longest time. Their GPUOpen initiative saw regular open-source contributions to a wide variety of projects. Every new feature that AMD works on can be found on the GPUOpen website. Moreover, AMD features tend to not be vendor-locked or generation-locked. Take, for example, FSR (FidelityFX Super Resolution) which was released a few years ago but was compatible with NVIDIA cards as well as AMD cards going back to the RX 480 that was launched in 2016. NVIDIA's latest Frame Generation technology only works on their latest RTX 40 series cards. That's because it relies on the Optical Flow Accelerator which is a hardware component that's present on RTX 40 cards. AMD's gaming drivers continue to have problems. It's not as bad as it used to be

but between the two, NVIDIA has the more stable set of drivers.

Then there's the case of encode streams. Until recently, NVIDIA had a software limitation of 3 streams on each graphics card which they recently increased to 5 streams in December 2023 and in January 2024 was increased to 8. AMD, on the other hand, never had any such artificial limits. NVIDIA's codec is better than AMD's but the artificial limit makes it not as sweet a deal as it could have been.

VALUE FOR MONEY

AMD excels in this department for multiple reasons. AMD has a more consumer-friendly approach to its hardware pricing and the underlying philosophy of supporting open-source initiatives. We've already touched upon how the technologies are compatible with older hardware and not necessarily locked to the very latest generation. Drivers improve performance over time for both, AMD and NVIDIA, but AMD tends to get a bit more performance as time goes by. Looking at the performance-per-rupee metrics, AMD cards are cheaper and with power consumption being very similar, even the running costs are comparable to NVIDIA. There was a time when AMD cards were known for being power sinks but those days are far behind. At the end of the day, an AMD card will work on more operating systems, last you for a long time thanks to upscaling technologies such as FSR and running costs will be similar to NVIDIA.

WHICH ONE IS IT, DOC?

Whatever fits your needs. We know that it feels like a cop-out but think of it this way, if you like ray tracing in your games, you would want to get an NVIDIA card. And if you primarily like to work on Linux, then an AMD card is what you should be looking to get. At the end of the day, these are two companies that innovate endlessly and come out with a lot of interesting technologies. And publications like ours test the latest and greatest to let you know which is better than the other. So, all you need to do is read our reviews and let us guide you. 